



You Only Grasp Once: Human-Inspired Benign Grasping for Agricultural Products

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Abstract

This study presents the development and evaluation of a multi-fingered, biomimetic robotic gripper. This gripper is designed to handle delicate, soft agricultural produce. To address this, an eye-in-hand vision system supports a multi-view look-and-grasp strategy, capturing object geometry from several angles to assess optimal grasp planning. Besides, a digital twin environment is used to simulate and evaluate candidate grasp configurations. The most stable configuration is validated in a physical setup using a Universal Robots UR3 arm. Results demonstrate the gripper's ability to achieve steady, damage-free grasps on strawberries as well as its versatility across various objects, including ping pong balls, plush toys, sushi, and Takoyaki. By showing how hybrid actuation and digital simulation can enhance robotic harvesting reliability and adaptability, this work advances precision agricultural robotics contributing to reduced post-harvest losses and improved food security in alignment with UN sustainable development goal 2 (zero hunger).

Keywords Robotic gripper · Precision agriculture · Hybrid actuation · Digital twin simulation

1 Introduction

Grasping is a fundamental capability in robotic manipulation, enabling a robot to interact physically with objects in its environment [1, 2]. A typical grasping operation consists of several stages: object perception, grasp planning, grasp execution, grasp holding, and object release [3–5]. Each of these stages requires careful coordination between sensing, decision-making, and actuation. Successful grasping depends not only on the mechanical design of the end-effector, but also on the quality of perception and the robustness of planning and control under uncertainties. In unstructured settings such as agriculture, where object sizes and geometries are

unpredictable, the variability in objects and environments make grasping particularly difficult [6–8].

The complexity of grasping increases significantly when target objects are fragile and easily damaged, as is the case with various agricultural products [9–11]. Fruit, such as strawberries, peaches, and tomatoes have soft and non-uniform surfaces, with irregular geometries and high sensitivity to compression. Excessive or uneven contact forces during grasping can lead to surface bruising, deformation, or even detachment of stems, thereby reducing product quality and shelf life [12–14]. These risks are further amplified in real-world agricultural scenarios, where lighting conditions vary, objects are clustered or occluded by leaves, and motion dynamics are unpredictable (Fig. 1).

In complex environments, humans can grasp delicate objects with ease and reliability. This ability relies on a combination of visual perception, native intelligence, motor coordination, and tactile feedback [15–17]. Humans naturally observe objects from multiple viewpoints, identify salient features, infer grasp affordances, and select hand postures that conform to the object's shape. By applying gentle and adaptive interaction, this process is fast, flexible, and context-aware. Enabled by fine motor control, such actions demonstrate a tight coupling between perceptual understanding

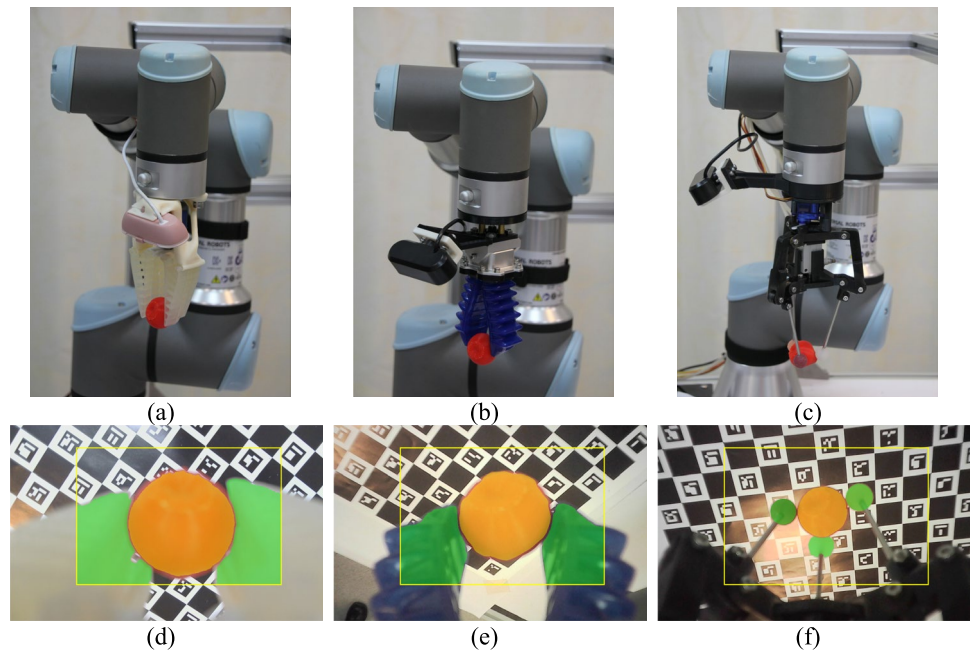
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Fig. 1 Comparison of gripper designs and their corresponding camera-frame occlusions: (a) Fin-ray gripper; (b) Pneumatic gripper; (c) Mini BarrettHand gripper; (d) Camera-frame occlusion caused by the Fin-ray gripper; (e) Camera-frame occlusion caused by the Pneumatic gripper; and (f) Camera-frame occlusion caused by the Mini BarrettHand gripper



and physical execution that current robotic systems cannot fully replicate [18, 19]. Imitating such human capabilities in robotic systems requires not only compliant hardware but also perceptual intelligence and predictive control strategies [20, 21].

In our research on robotic grasping for agriculture, we have mainly focused on the development of various compliant and underactuated grippers, including Fin-ray [22] and pneumatic grippers [23]. Despite the availability of such new gripper designs, current solutions fail to achieve the delicate balance between secure grasping and damage avoidance [24, 25]. Consequently, in precision agriculture, the automation of fruit harvesting remains an unresolved challenge [26, 27]. To address this gap, we introduce a human-inspired compliant gripper capable of adapting to variability in object geometry and pose. The proposed design demonstrates improved grasp reliability and adaptability. Its efficacy and adaptability contribute to better handling of products, thereby reducing food wastage and damage. This study presents a human-inspired robotic grasping framework designed specifically for handling delicate agricultural products.

2 Gripper Design and Control

This section presents the integrated mechanical, electrical, and computational foundations of the proposed gripper designed for strawberry harvesting. To accommodate the irregular shapes and delicate surfaces of soft agricultural produce, the gripper architecture emphasizes structural simplicity, modularity, and compliance. A hybrid actuation

system, comprising a pneumatic actuator for coordinated grasping and servo motors for independent finger orientation, enables adaptive and damage-free manipulation. An eye-in-hand vision system is embedded to support perception-driven control. This system allows the gripper to interpret object geometry and adjust grasp strategies in real-time. Complementing the physical design, a digital twin simulation environment developed in CoppeliaSim promotes pre-execution testing, performance optimization, and iterative refinement of control parameters. Together, these components form a cohesive and scalable framework that addresses the key challenges of soft object handling in unstructured agricultural environments.

2.1 Mechanical Design and Fabrication

As shown in Fig. 2, the proposed gripper is a compact, lightweight, and modular end-effector specifically designed for the delicate harvesting of strawberries and other soft agricultural produce. Inspired by biomimetic multi-fingered grippers with redundant joints, the design prioritizes adaptability, compliance, and straightforward integration with existing agricultural robotic platforms. While prior approaches leveraging underactuated mechanisms have demonstrated effective adaptive grasping, they often introduce mechanical and control complexity. In contrast, the present design adopts a simplified yet functionally dexterous architecture that minimizes the number of actuators and reduces the mechanical footprint.

An eye-in-hand camera is mounted directly at the wrist of the gripper, providing a consistent visual perspective throughout the approach phase. This configuration

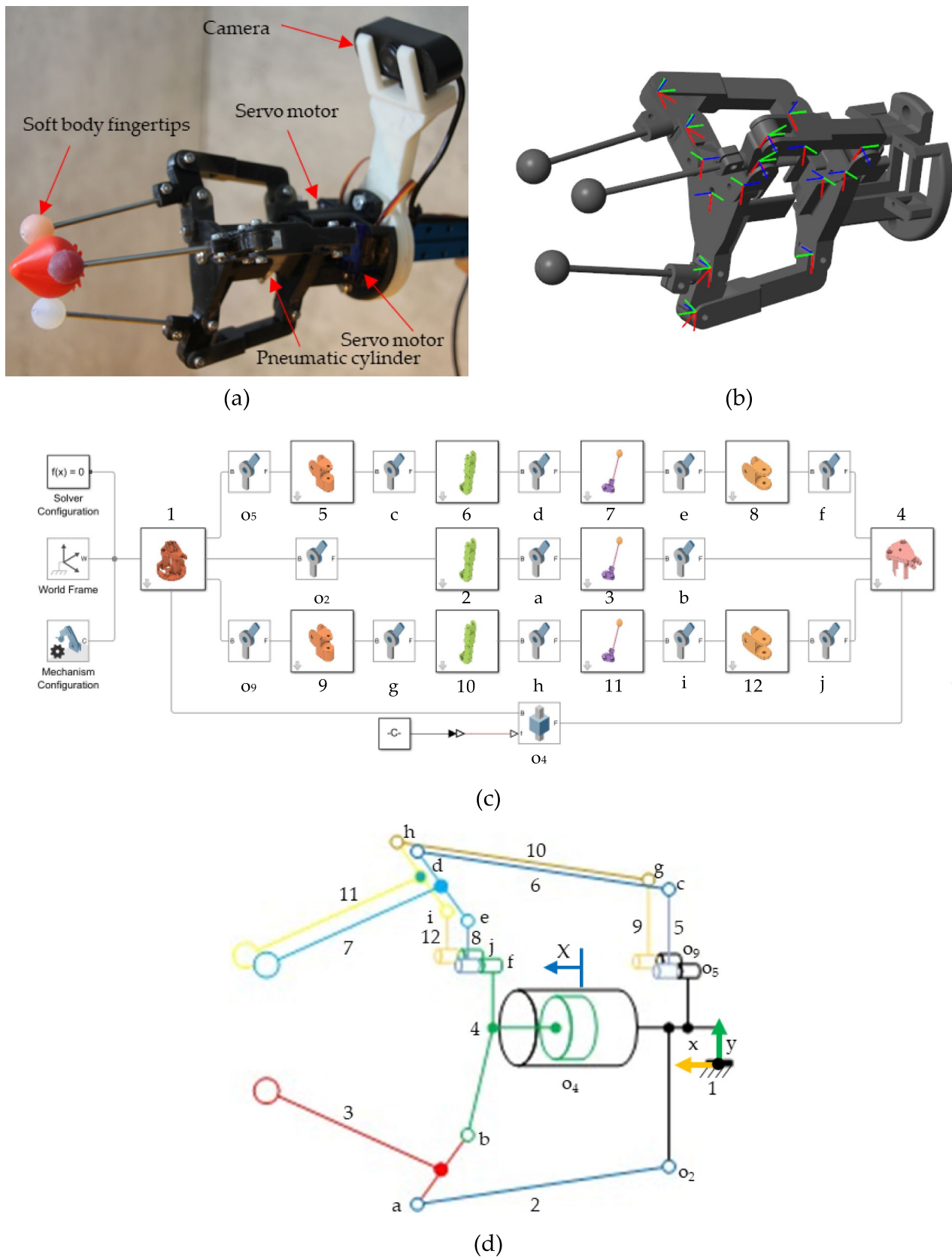


Fig. 2 A mini BarrettHand gripper architecture: (a) Photograph; (b) CAD model; (c) Simulink diagram; and (d) Kinematic diagram

supports real-time, vision-based control strategies, including object localization, pose estimation, and multi-view grasp planning. Visual data is processed externally on a notebook computer, which performs high-level perception and planning tasks and communicates actuation parameters to the gripper. Compared to the occlusion rates of 36.76% and 29.73% observed using a conventional Fin-ray (Fig. 1d) and pneumatic gripper (Fig. 1e), respectively, the proposed design achieves a significantly lower occlusion rate of 5.09% (Fig. 1f), thereby enhancing the visibility of the camera during grasp planning.

Our gripper comprises three fingers, each consisting of two links connected by two joints, resulting in a total of six links and six joints (Fig. 2d). Three revolute joints facilitate finger rotation and flexion, while three cylindrical joints, implemented via a shared pneumatic actuator, enable synchronized grasping. A single double-acting pneumatic cylinder, actuated through a 5/2 solenoid valve, drives a common linkage that simultaneously actuates all three fingers. This configuration provides uniform, pressure-regulated grasping force across the fingers, allowing compliant interaction with delicate surfaces and reducing the risk of bruising. The grasping motion itself is generated through an inverse slider–crank mechanism, where the pneumatic cylinder actuates the slider; as the cylinder retracts, it rotates the crank, causing the soft grasping balls attached to the crank to move inward toward the object and execute the grasping action.

To enhance adaptability, two servo motors independently control the orientation of the outer fingers via base-mounted rotational joints. The workspace and rotation range of each finger is shown in Table 1. These redundant degrees of freedom (DOFs) support pre-grasp adjustments and allow the gripper to adapt to varying fruit positions and geometries, particularly in cluttered or irregular environments. Operated in position control mode, the servo-driven joints, together with the pneumatic actuator, realize a total of three active DOFs: one for synchronized grasping and the other two for finger orientation.

The gripper's structural components are fabricated using 3D printing, facilitating rapid prototyping, low-cost iteration, and straightforward replacement. Designed for compatibility with Universal Robots (UR) arms, the gripper features a standard wrist flange for plug-and-play integration. By combining mechanical compliance, hybrid actuation, and vision-guided control, the proposed design

offers a reliable and scalable solution for precision harvesting in agricultural robotics.

2.2 Microprocessor, Electronics, and Pneumatic Control

The electronic control architecture of the gripper is centered on a Raspberry Pi Pico board, a compact and energy-efficient microcontroller. It functions as the interface between high-level control algorithms and low-level actuation hardware (Fig. 3). The Pico board generates pulse-width modulation (PWM) signals to drive two servo motors, which independently adjust the orientation of the outer fingers. Operating in position control mode, these motors enable real-time reconfiguration of the gripper to accommodate variations in fruit pose and spatial arrangement.

The primary grasping motion is actuated by a double-acting pneumatic cylinder, which is controlled by a 5/2 solenoid valve. Both the 24-V supply for the valve and the 5-V supply for the servo motors are sourced from the regulated outputs of the robot controller, ensuring centralized power distribution and seamless synchronization with the robotic system. Acting as a logic-level controller, the Pico board generates digital output signals that activate a relay module for switching the solenoid valve. This arrangement provides electrical isolation between the low-voltage logic circuitry and the high-voltage pneumatic components.

The Pico board manages timing and actuation at the embedded level, while high-level perception and planning are performed on an external notebook computer, which receives real-time imagery from the eye-in-hand camera for object detection and pose estimation. The resulting control parameters are transmitted to the Pico board via USB or UART, maintaining a clear separation between high-level computation and low-level actuation. This distributed architecture promotes modularity, while minimizing computational overhead on the embedded platform. It also establishes a flexible foundation for future enhancements, such as learning-based control or closed-loop feedback integration.

Table 1 Description of workspace and rotation range of each finger

Finger	Range [degree]	γ_i [degree]
Finger 1	120	[0, 120]
Finger 2	120	[0, -120]

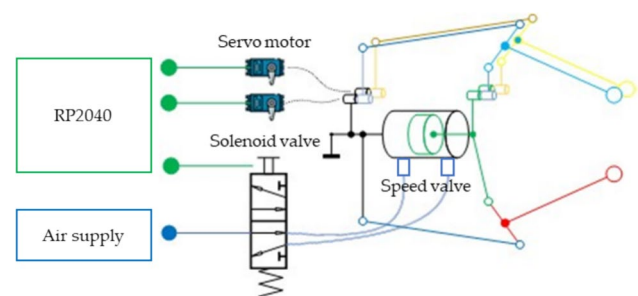


Fig. 3 Electrical and pneumatic control systems of the proposed gripper

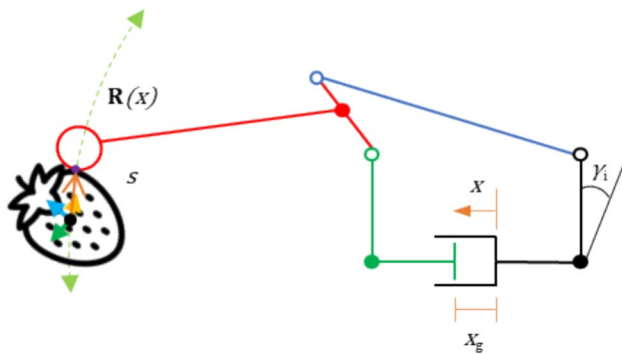


Fig. 4 Static-force and kinematic diagram of the proposed gripper

For testing and calibration, a manual push-button is integrated into the circuit to allow on-demand actuation. A pressure regulator is also incorporated into the pneumatic line to control the speed of cylinder motion, thus facilitating smooth and stable grasping under varying mechanical loads (Fig. 4).

2.3 Mathematical Model of the Gripper (Table 2)

The gripper design is inspired by the BarrettHand architecture [30, 31], and employs three inverted slider-crank mechanisms to generate synchronized grasping motion [32, 33]. These mechanisms are mechanically coupled via a shared slider linkage, so that translating the slider simultaneously actuates all three fingers. To enable independent control of finger abduction without interfering the primary grasping motion, two of the slider-crank mechanisms are mounted on rotating joints aligned with their axes of rotation. This configuration allows for adjustable finger spread through servo-driven rotation of the outer fingers. The gripper features three actuation inputs: one translational DOF, denoted as x , corresponding to the linear displacement of the pneumatic cylinder, and two angular DOFs, represented by γ_1 and γ_2 , which control the abduction angles of the first and second fingers, respectively. The forward

kinematics of each fingertip T_{0i} relative to the base frame can be expressed as a function of these inputs:

$$\begin{bmatrix} T_{01} \\ T_{02} \\ T_{03} \end{bmatrix} = \begin{bmatrix} f(x) \cdot T_{1,r}(\gamma_1) \cdot T_1 \\ f(x) \cdot T_{2,r}(\gamma_2) \cdot T_2 \\ f(x) \cdot T_3 \end{bmatrix} \quad (1)$$

where $f(x)$ denotes the transformation resulting from the slider-crank motion induced by the pneumatic displacement x . The term $T_{i,r}(\gamma_i)$ represents the rotation matrix corresponding to the abduction angle γ_i for fingers 1 and 2. Variable T_i is the fixed transformation matrix from the base of the gripper to the initial frame of finger i . The resulting fingertip pose with respect to the base frame is denoted by T_{0i} . In Table 1, the descriptions of all variables in this study are listed.

Let $q = [x, \gamma_1, \gamma_2]^T$ denote the generalized coordinates of the gripper, and $\tau_{act} = [P, \tau_1, \tau_2]^T$ the corresponding generalized efforts. By differentiating (1), the fingertip displacement δr_i is related to the joint displacement δq through the fingertip Jacobian:

$$\delta r_i = J_i(q) \delta q \quad (2)$$

The grasping contact forces g_i can then be related to the generalized actuation input through the principle of virtual work. For any virtual displacement δq , static equilibrium requires

$$\tau_{act}^T \delta q + \sum_{i=1}^3 g_i^T (J_i(q) \delta q) = 0 \quad (3)$$

where τ_{act} represents the generalized actuator effort and $J_i(q)$ is the fingertip Jacobian of finger i .

Since contact with the object constrains the pneumatic cylinder position x , which itself depends on the object shape s , the relative pose T_{og} , and the abduction angles γ_1 and γ_2 , the magnitude $|g_i|$ can thus be written as:

$$|g_i| = f(q, \tau_{act}, s, T_{og}) \quad (4)$$

This function reflects the combined influence of input pressure, object geometry, relative alignment, and abduction configuration on the force delivered at the contact point. Equation (4) considers only the component of grasping force aligned with the motion path of the fingertips.

3 Vision-Guided Planning and Rehearsal Simulation

3.1 Grasp Planning Strategy

The proposed grasp-planning strategy integrates multi-view visual perception with a compact, task-oriented grasp library to enable reliable and benign handling of delicate agricultural products, such as strawberries. The planning process

Table 2 Description of variables used in the grasping system model

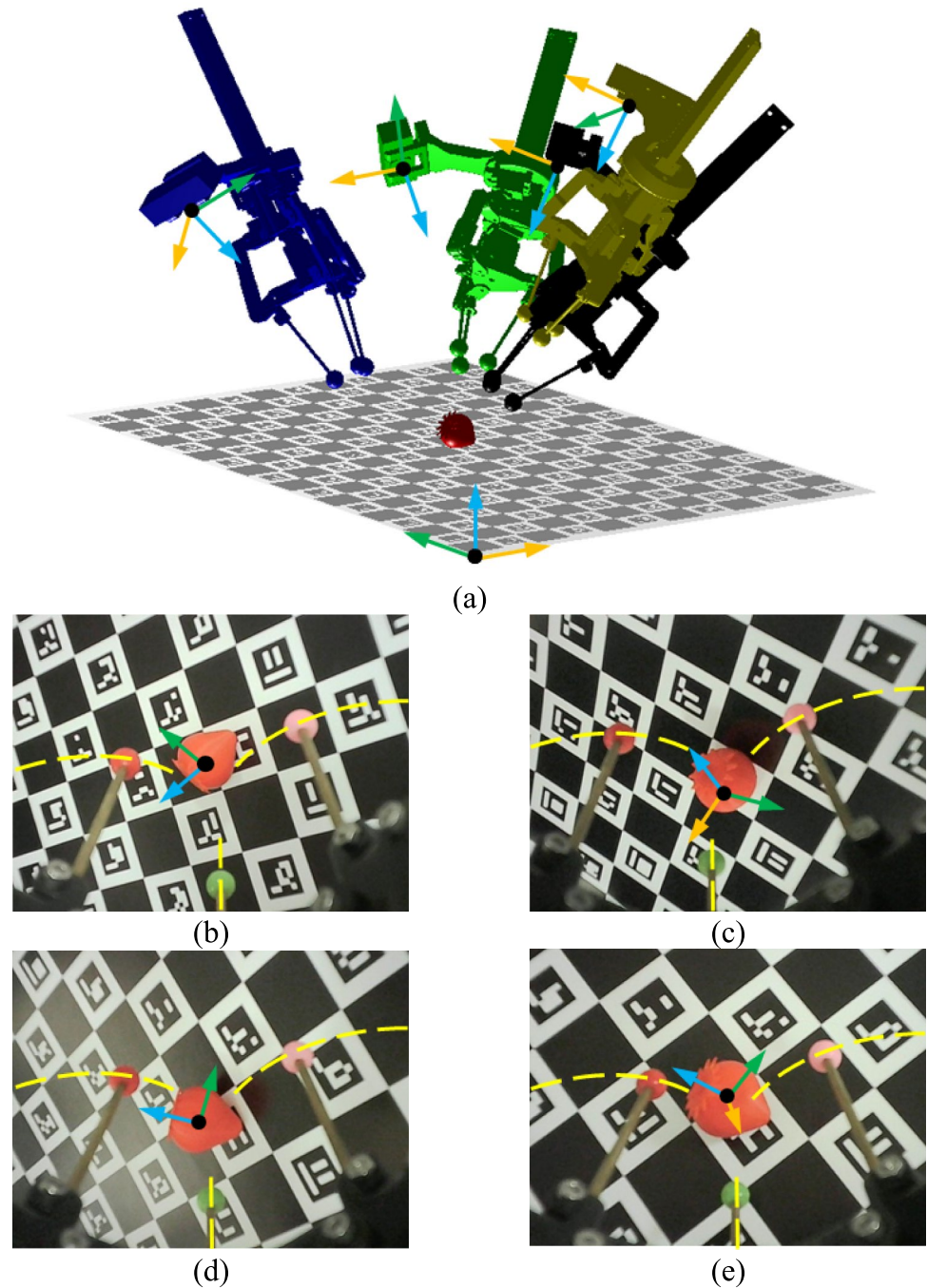
Variable	Meaning
x	Linear actuation input
x_g	Pneumatic cylinder grasping position
γ_i	Angle of finger i
$f(x)$	Finger tips transformation
s	Object geometry
P	Applied pressure
g_i	Contact force on finger i
r_i	Motion path of finger i
T_{ij}	Transformation matrix of frame j with respect to frame i
T_i	Transformation matrix from base to initial frame of finger i
J_i	fingertip Jacobian of finger i

begins with a multi-angle observation phase, during which the gripper, equipped with an eye-in-hand camera, inspects the target object from multiple viewpoints (Fig. 5). These views are used to analyze the object's geometry, orientation, and occlusion characteristics, providing robust perceptual inputs for subsequent grasp selection.

Using aggregated information from the multi-view observations, the system selects an appropriate grasp configuration from a predefined library of motion primitives. The grasp

library is deliberately designed as a small and interpretable set of canonical grasp modes, reflecting the primary functional capabilities of the proposed gripper rather than attempting to enumerate an exhaustive grasp taxonomy. Each grasp primitive specifies the intended finger configuration, end-effector approach direction, and nominal actuation parameters. In the current implementation, the library includes parallel grasp, three-finger pinch, asymmetric adaptive grasp, as well as open and release configurations.

Fig. 5 Multi-view visual perception of a target strawberry: (a) Observation poses; (b) Front view; (c) Top view; (d) Left-side view; (e) Right-side view; and (f) Multiple view strategies for grasping



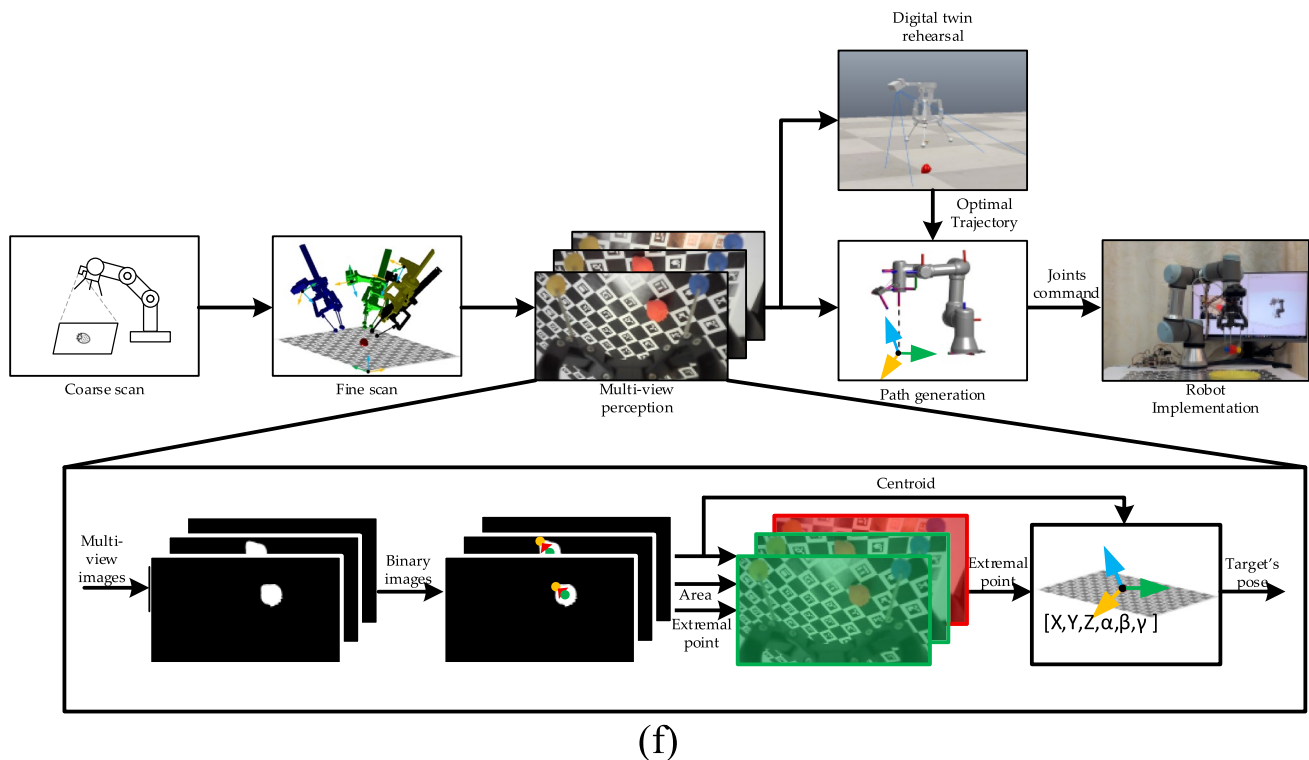


Fig. 5 (continued)

Among these, the three-finger pinch is the primary grasp mode employed in the experimental evaluation.

Grasp selection is implemented using a rule-based strategy driven by geometric cues extracted from multi-view perception, such as object centroid location, visible surface distribution, and approach feasibility under partial occlusion. This design choice prioritizes transparency, robustness, and computational efficiency, while avoiding reliance on data-intensive learning-based affordance reasoning. Rather than increasing the number of grasp types, the framework leverages low camera occlusion and improved multi-view visibility to enhance confidence when selecting among a limited set of physically meaningful grasp primitives.

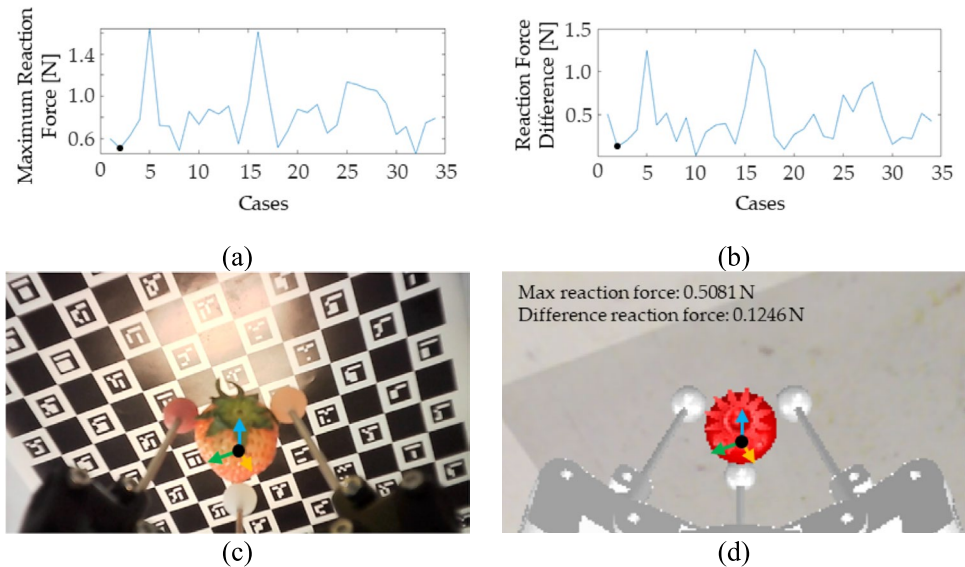
Once a target grasp is selected, the system transitions to the execution stage. Candidate approach trajectories are first evaluated within a digital twin environment to estimate grasp feasibility and force distribution under quasi-static conditions. Following this evaluation, the selected grasp primitive is executed on a UR3 robotic arm using URScript-based command templates. Each motion primitive is mapped to a parameterized URScript routine that combines robot motion commands (e.g., `movej`, `movel`) with gripper actuation signals for pneumatic and servo control. The grasp planner instantiates these routines by inserting the target end-effector pose and finger configuration derived from perception, enabling concise, repeatable, and reliable execution on the robot controller.

This two-stage pipeline—perceptual reasoning and simulation-assisted planning followed by a single physical grasp attempt—forms the core of the proposed vision-guided grasping framework and directly supports the You Only Grasp Once (YOGO) principle. Although the grasp library is fixed in the present study, it is extensible by design. Future work may incorporate simulation-generated datasets and learning-based selection mechanisms to expand or refine the library, enabling adaptation to new object geometries while preserving the system’s perception-aware and planning-centric framework.

3.2 Grasp Position and Force Analysis

To predict internal physical dynamics of the grasp, such as reaction forces and grasp stability, which are not directly observable through the camera, the proposed system incorporates a digital twin framework. As illustrated in Fig. 6, the digital twin simulates the contact interactions between the gripper fingers and the strawberry, providing real-time estimates of critical metrics, including the maximum reaction force and force imbalance across contact points. These simulated force profiles enable the system to assess whether a grasp is stable and gentle enough for the delicate fruit, even before physical contact occurs. For instance, a predicted force imbalance may indicate an off-centered grasp or uneven finger engagement, prompting the system to adjust finger trajectories or reconfigure its approach accordingly. During the planning phase, the system’s predictive

Fig. 6 Grasping outcome evaluation for the proposed gripper: (a) Reaction force distribution at maximum grasp force; (b) Force variation between maximum and minimum grasp conditions; (c) Photograph of a real-world grasping trial; and (d) Corresponding simulated grasp in the digital-twin environment



capability extends situational awareness into the mechanical domain, thereby facilitating more informed decision-making. This capability enhances operational robustness and mitigates the risk of fruit damage, particularly in dynamic or partially occluded harvesting environments.

3.3 Grasping Rehearsal in the Digital Twin Environment

The proposed gripper was evaluated via a virtual simulation environment using a digital-twin framework. As depicted in Fig. 7a, the simulation accurately replicates the gripper's kinematics, actuation behavior, and interactions with deformable objects such as strawberries. Within this virtual

environment, the system can test and refine grasping strategies under controlled variations in object position, orientation, and compliance: namely, softness. The digital twin enables the autonomous evaluation of candidate grasp configurations by simulating finger contact forces and predicting grasp stability, without physical execution (Fig. 8b). This capability is essential for identifying robust motor strategies, improving task success rates, and reducing the need for real-world trial-and-error. Moreover, the digital twin supports parameter tuning and failure analysis by allowing the comparison of simulated outcomes across different motion trajectories and grasp geometries. Transferring iterative learning to a safe virtual environment reduces hardware fatigue and enhances the system's adaptability. The resulting

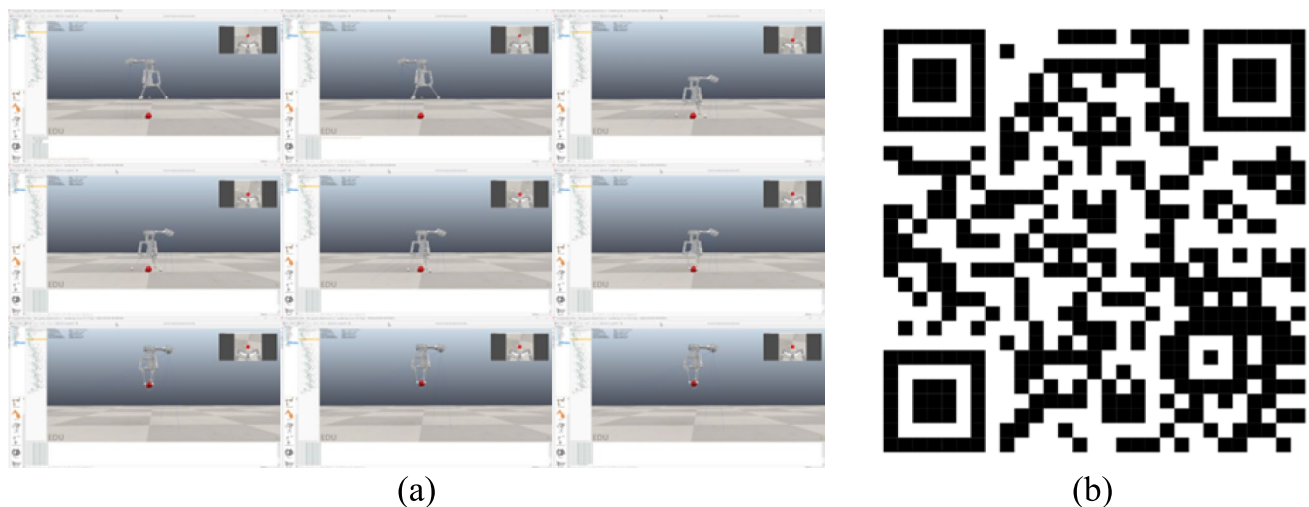
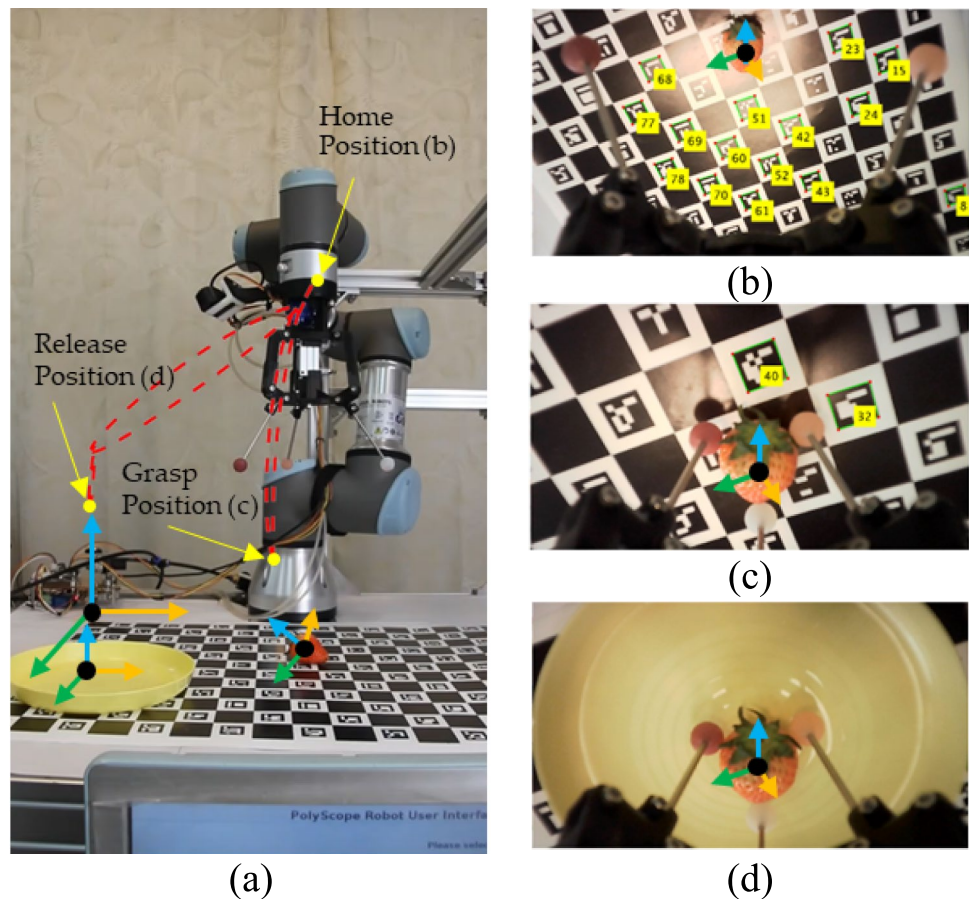


Fig. 7 Digital-twin simulation results: (a) Time-lapse of the grasp-planning process; and (b) Visualization of grasp execution in simulation

Fig. 8 Gripper trajectory during a pick-and-place task: **(a)** Full trajectory; **(b)** Start position; **(c)** Grasping position; and **(d)** Object placement position



insights are directly applicable to the physical robot, effectively shortening development cycles and improving real-world performance.

4 Results and Discussions

4.1 Real-World Grasping Trials

To assess the practical performance of the proposed gripper, a physical grasping experiment was conducted using a UR3 robot platform. The experiment involved autonomously identifying, grasping, and relocating a strawberry from the floor to the center of a yellow plate (Fig. 8a). The robot initiated the sequence by positioning the gripper module above the strawberry, enabling a top-down view via the integrated eye-in-hand camera (Fig. 8b). To detect the strawberry, estimate its pose, and determine an optimal grasp configuration, computer vision techniques were applied. Subsequently, the robot followed a predefined motion trajectory to approach and grasp the strawberry using the multimodal gripper (Fig. 8c). Once secured, the gripper transported the strawberry to the target plate and carefully released it at the designated location (Fig. 8d).

The entire manipulation sequence was recorded for post-experiment analysis. Although video data were processed offline in this study, the system architecture is designed to accommodate future integration of real-time visual feedback, closed-loop control, and adaptive grasping, enabled by ongoing advances in onboard computation and artificial intelligence. This experiment serves as a proof of concept, demonstrating the feasibility of the proposed grasping system under semi-structured conditions. The successful execution highlights the gripper's compliance, vision-guided planning capabilities, and its potential applicability in automated agricultural harvesting.

4.2 Evaluation of Grasp Quality

Grasp stability was evaluated using visual feedback from an eye-in-hand camera, which continuously tracked the positions of the three fingertips relative to the grasped object. This setup enabled a frame-by-frame analysis of the spatial consistency between the gripper and the target, providing a quantitative measure of grasp stability throughout the manipulation task (Fig. 9a-d). By monitoring the relative motion between the fingers and the strawberry across various trajectories, the mechanical fidelity and control accuracy

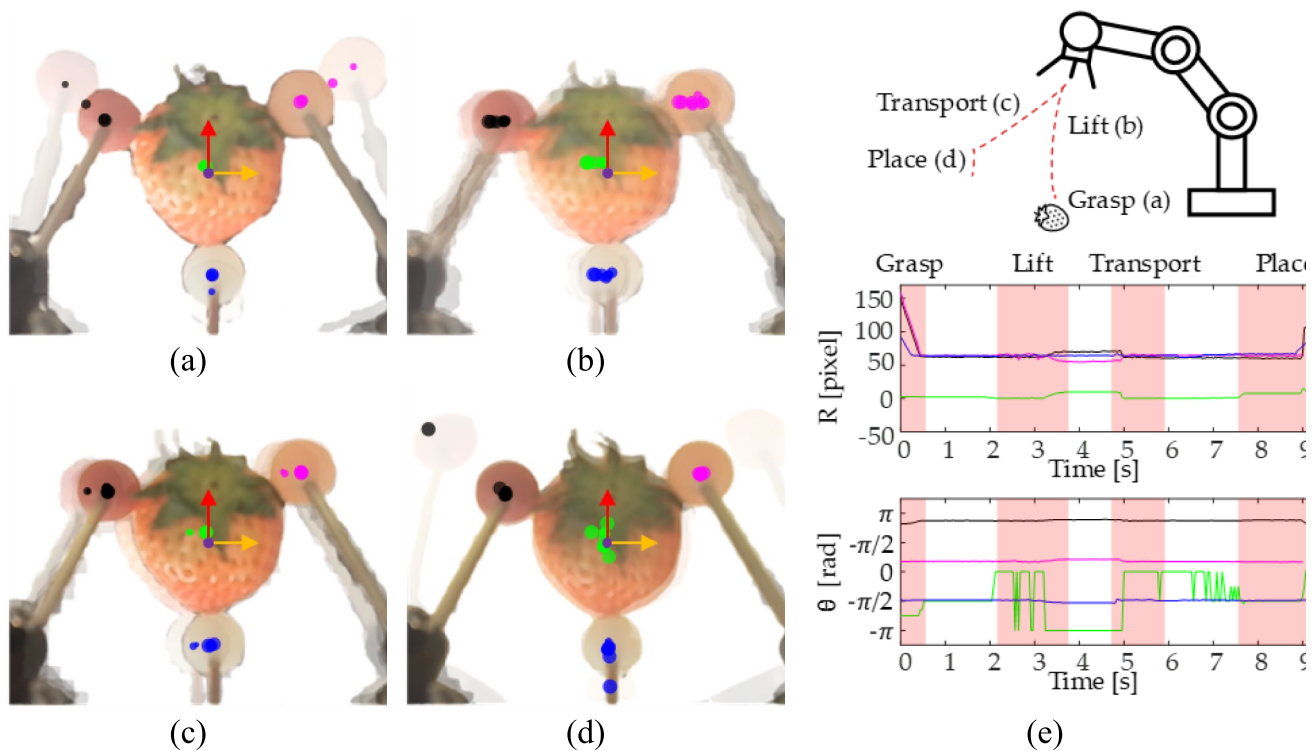


Fig. 9 Evaluation of grasping performance using an eye-in-hand camera: (a) Grasp; (b) Lift; (c) Transport; (d) Place; and (e) Distance and angle relative to strawberry center over time

of the design were assessed (Fig. 9e). The analysis indicated that the gripper maintained consistent contact and positional alignment in most scenarios, reflecting effective force distribution and adaptive compliance. However, during execution of the Lift trajectory, a relatively fast and curved path, noticeable oscillations occurred between the fingertips and the object. These oscillations were observed as transient deviations in fingertip alignment, suggesting minor instabilities likely caused by pneumatic compliance and inertia effects during high-dynamic motion. The ability to detect such instabilities through onboard vision underscores the system's potential for closed-loop feedback integration and real-time corrective control in future implementations. Overall, the results confirm that the gripper achieved stable and responsive grasping under moderate dynamic conditions, while revealing opportunities for further improvement in active damping and trajectory-smoothing strategies.

4.3 Versatility in Grasping Similar Objects

To assess the generalizability of the proposed gripper beyond strawberry harvesting, physical grasping experiments were conducted using objects of similar size but different shapes, textures, and mechanical properties. The objective was to validate the gripper's adaptability in handling diverse items,

demonstrating practical challenges encountered in both agricultural and service robotics. Five objects were selected for this study:

- Strawberry: baseline reference object (Fig. 10a),
- Ping-pong ball: lightweight, rigid, and smooth (Fig. 10b),
- Toy doll: soft, irregularly shaped with nonuniform geometry (Fig. 10c),
- Takoyaki (octopus ball): moderately compliant with slight surface adhesiveness (Fig. 10d),
- Sushi: fragile and deformable with low structural cohesion (Fig. 10e).

In all experimental cases, the gripper achieved stable grasps without visible deformation or slippage, demonstrating its capability to handle a wide range of surface textures and mechanical characteristics. The integration of pneumatic compliance with servo-controlled finger orientation enabled effective adaptation to different contact profiles and object geometries. These results highlight the versatility of the gripper design and its potential for broader application in delicate object handling. All trials were completed without any hardware modifications or control parameter tuning, underscoring the system's robustness and general-purpose utility.

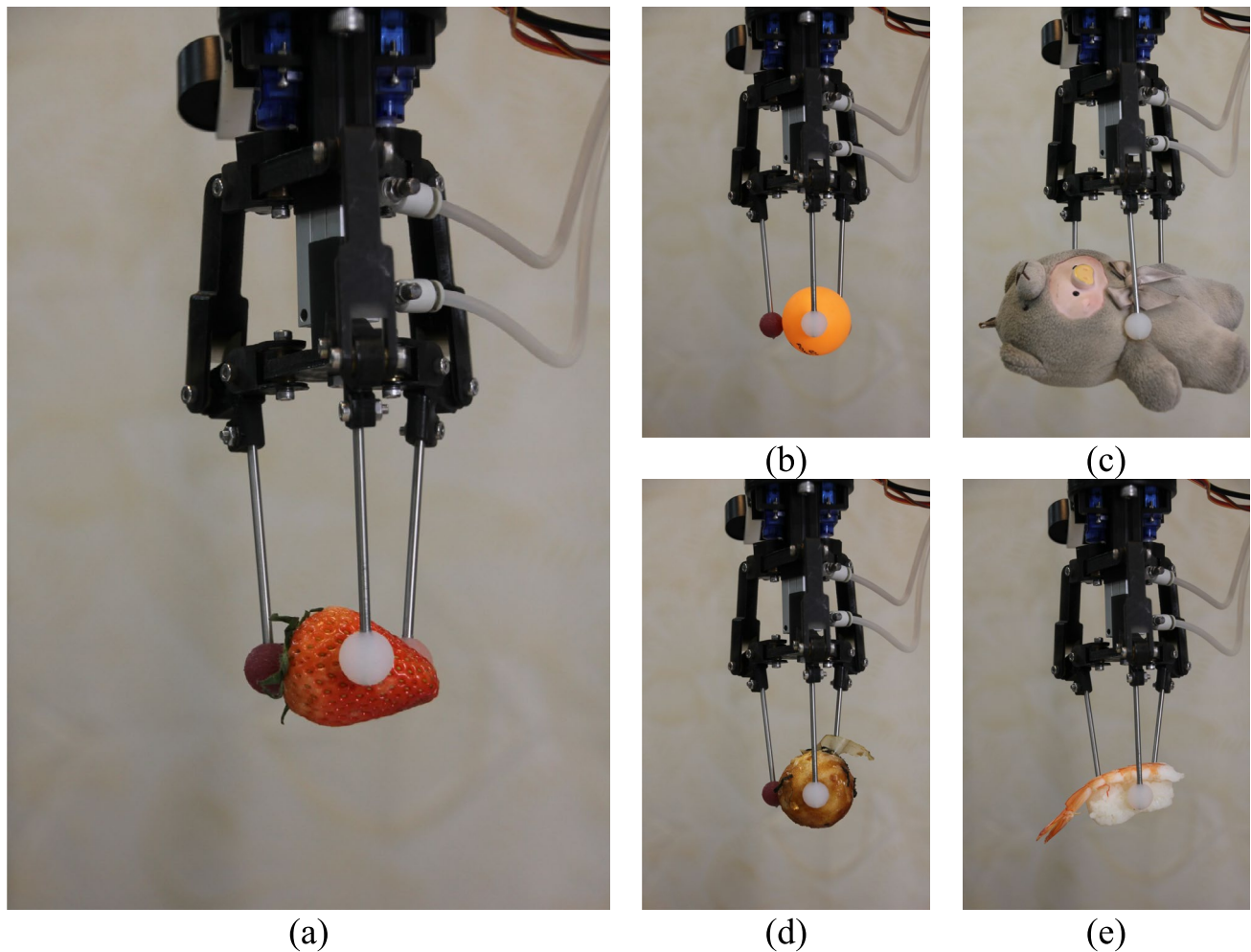


Fig. 10 Grasping demonstration on various objects using the proposed gripper: (a) Strawberry; (b) Ping-pong ball; (c) Toy doll; (d) Takoyaki (octopus ball); and (e) Sushi

4.4 Digital Twin Force Validation

To assess the validity of the grasping force estimated by the digital twin, an experimental force measurement was carried out using the proposed gripper. As illustrated in Fig. 11(a), the setup consists of the gripper instrumented with three strain-gauge sensors, each measuring the contact force at an individual fingertip. All experiments were conducted under controlled pneumatic pressure to ensure repeatable and quasi-static grasping conditions.

During the experiment, the supplied pneumatic pressure was varied incrementally, and the resulting fingertip forces were recorded from the strain-gauge sensors. These measurements were then compared with forces predicted by the digital twin model implemented using a MATLAB multibody representation of the gripper, evaluated under identical actuation conditions. The comparison between

measured and modeled forces is shown in Fig. 12(b). All validation experiments were performed using the three-finger pinch grasp configuration, which is the primary grasp mode evaluated in this study.

The results indicate a strong correspondence between experimental measurements and simulation predictions across the tested pressure range, demonstrating that the digital twin captures the dominant force-generation behavior of the gripper. While minor discrepancies are observed, these are attributed to effects such as joint friction and material compliance that are intentionally neglected to preserve model simplicity and computational efficiency.

Overall, the experimental results confirm that the proposed digital twin provides sufficiently accurate static force estimation for its intended role in open-loop grasp planning and pre-execution evaluation, supporting its use within the proposed You Only Grasp Once (YOGO) framework.

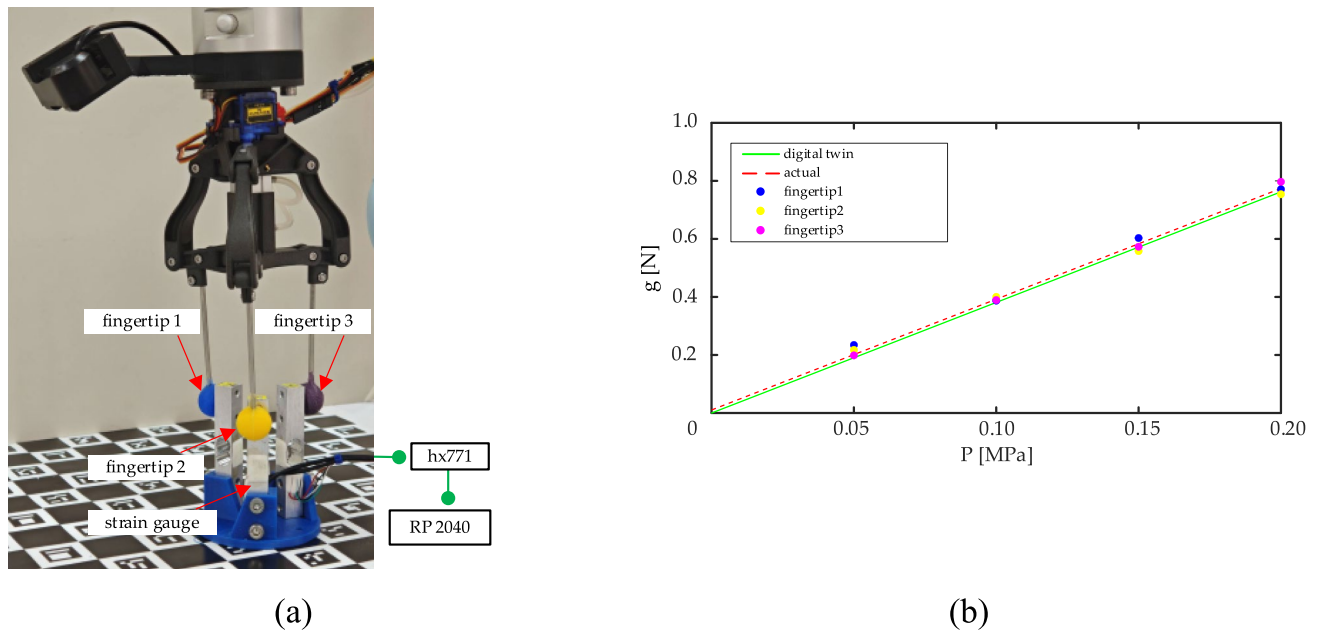
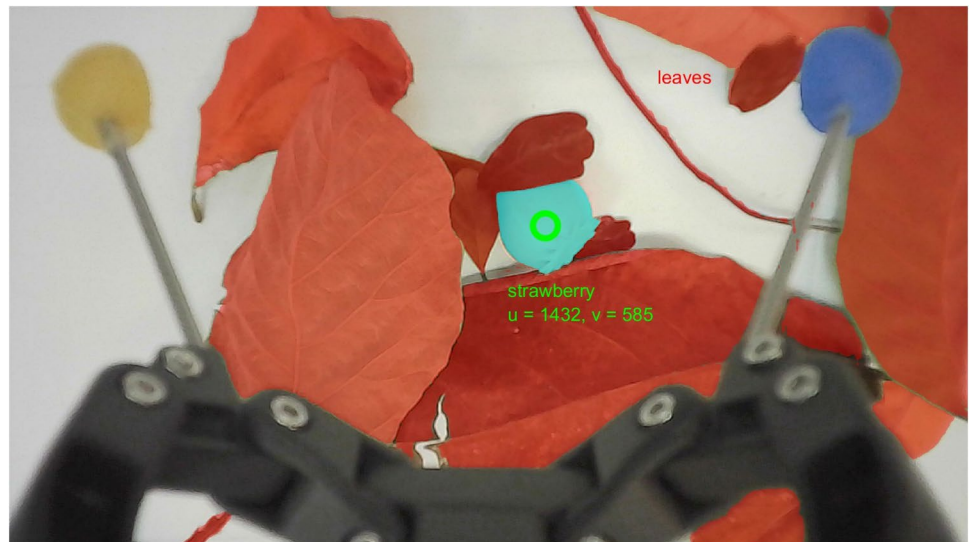


Fig. 11 Experimental validation of grasping force. (a) Experimental setup consisting of the proposed gripper and three strain-gauge sensors. (b) Comparison between the measured and modeled grasping force

Fig. 12 Image analysis in a cluttered environment



4.5 Practical Applicability and Limitations

Agricultural environments are inherently cluttered, with target fruits often partially occluded by leaves, stems, or neighboring objects. To evaluate system robustness under such conditions, the proposed framework incorporates image analysis capable of operating in visually complex scenes.

At this stage, standard image processing and deep-learning-based semantic segmentation techniques are employed to identify the target fruit and extract basic spatial information. While these vision methods enable reliable object

detection and localization, advanced segmentation or learning-based perception is not the primary focus of this work. Instead, the emphasis is placed on how the proposed gripper design and multi-view observation strategy support grasp planning despite partial occlusions.

Figure 12 illustrates representative camera views obtained using the proposed eye-in-hand gripper assembly in a cluttered scene, along with the resulting positional analysis of the strawberry. The results demonstrate that even when parts of the fruit are occluded in individual views, sufficient visual information can be obtained to support grasp planning.

The proposed system is designed as a planning-centric grasping framework for delicate agricultural products, emphasizing reliable grasp acquisition through perception-driven evaluation rather than dynamic manipulation. Its compact, lightweight gripper design and eye-in-hand configuration make it suitable for deployment on small industrial manipulators commonly used in agricultural automation, such as the UR3 platform demonstrated in this study.

However, several limitations must be acknowledged. First, the system is currently designed to operate on a single object of interest at a time, and does not address simultaneous grasp selection in dense clusters of overlapping fruits. Second, as the perception pipeline relies primarily on camera-based image processing, variability in illumination conditions, shadows, or strong reflections may degrade segmentation accuracy, particularly in outdoor or unstructured environments. Third, the grasp planning assumes that the target object remains quasi-static during multi-view observation. As such, the system is not intended for scenarios involving strong wind, rapid object motion, or significant external disturbances during perception and execution. Finally, grasp execution is performed in an open-loop manner, and dynamic compensation during post-grasp manipulation lies outside the scope of this work.

Despite these limitations, the system provides a practical and robust foundation for perception-aware grasp planning in precision agriculture. Addressing dynamic environments, lighting robustness, and closed-loop manipulation are identified as important directions for future extension toward full field deployment.

5 Conclusion

In this study, a novel human-inspired grasping system was developed to address the challenges of delicate object manipulation in agricultural robotics. By integrating a compliant hybrid-actuated gripper with vision-guided planning and a digital twin simulation framework, the system enabled precise, adaptive, and damage-minimizing interactions with soft agricultural products. Results demonstrated that the proposed grasping platform could successfully enhance the reliability of robotic harvesting while minimizing product damage. Real-world trials further validated the gripper's versatility, enabling stable handling of diverse objects without hardware modifications or parameter tuning. The integration of predictive simulation and multimodal sensing provides a solid foundation for future research in real-time grasping control. This work contributes to the advancement of scalable and sustainable agricultural automation, aligning with the goals of precision farming and UN Sustainable Development Goal 2 (Zero Hunger).

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Author Contributions Data Availability Statement: All supplementary materials are publicly available in the project repository at <https://github.com/Poom1150/YouOnlyGraspOnce>.

Data Availability All supplementary materials are publicly available in the project repository at <https://github.com/Poom1150/YouOnlyGraspOnce>.

Declarations

Competing interests The authors declare no competing interests.

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